then the behavior of the other regions of the system is

$$\begin{aligned} x(t_1) &= F_0 x(0) + g_0 \\ x(t_2) &= F_1 x(1) + g_1 \\ &\vdots \\ x(t_N) &= F_{N-1} x(N-1) + g_{N-1} \end{aligned} \tag{2.6}$$

with

$$F_i = e^{A_{\sigma(t_i)}(t_{i+1}-t_i)} \tag{2.7}$$

$$g_i = \int_{t_i}^{t_{i+1}} e^{A_{\sigma(t_i)}(t_{i+1}-\tau)} b_{\sigma(t_i)} d\tau = \int_{0}^{t_{i+1}-t_i} e^{A_{\sigma(t_i)}\tau} b_{\sigma(t_i)} d\tau \tag{2.8}$$

From (2.6) we can establish the relation between $x(t_N)$ and $x(0)$

$$\begin{aligned} x(t_N) = {} & F_{N-1}F_{N-2}\dots F_0 x(0) + \\ & F_{N-1}F_{N-2}\dots F_1 g_0 + F_{N-1}F_{N-2}\dots F_2 g_1 + \dots + F_{N-1} g_{N-2} + g_{N-1} \\ x(t_N) = {} & F_N F_{N-1}\dots F_1 x(0) + c \end{aligned} \tag{2.9}$$

We know that $x(t_N) = x(0)$, so the solution for $x(0)$ is

$$\begin{aligned} x(0) &= F_{N-1}F_{N-2}\dots F_0 x(0) + c \\ \Big(I - F_{N-1}F_{N-2}\dots F_0\Big) x(0) &= c \\ x(0) &= \Big(I - F_{N-1}F_{N-2}\dots F_0\Big)^{-1} c \end{aligned} \tag{2.10}$$

where

$$c = F_{N-1}F_{N-2}\dots F_1 g_0 + F_{N-1}F_{N-2}\dots F_2 g_1 + \dots + F_{N-1} g_{N-2} + g_{N-1} \tag{2.11}$$

assuming that the inverse indicated in (2.10) exists.

2.3 Application Examples

2.3.1 Buck-boost converter

Considering a Buck-Boost converter in ~~a~~ Continuous Conduction Mode (CCM) whose topology is shown in Figure 2.2.

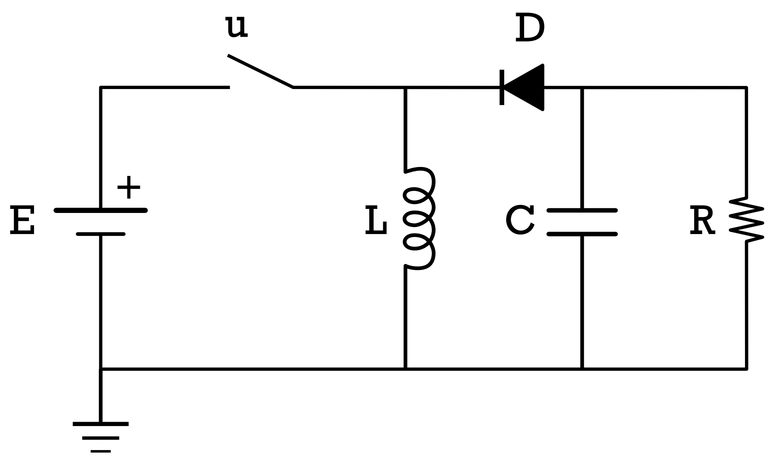

FIGURE 2.2 – Buck-Boost converter

The state variables are the capacitor voltage v_c and the inductor current i_L. Its dynamics are given by the equation:

$$\dot{x} = \begin{cases} A_1 x + b_1, \text{ for u} = 1 \text{ (closed switch - Mode 1)} \\ A_2 x + b_2, \text{ for u} = 0 \text{ (open switch - Mode 2)} \end{cases} \tag{2.12}$$

where

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & \frac{1}{\boxed{L_1}} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix}$$
$$b_1 = \begin{bmatrix} \frac{E}{L} & 0 \end{bmatrix}^T \qquad b_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}^T \tag{2.13}$$

where $R = L = C = E = 1$.

The list of dynamics for this example is

$$\Big\{ [A_1, b_1], [A_2, b_2] \Big\} \tag{2.14}$$

where

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tag{2.15}$$

$$A_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.16}$$

The criteria for the optimization is to compute the trajectory to minimize oscillation around the average value $[2, -1]$ with the weight matrix $Q = diag[1, 1]$, the maximum cycle period of $T_{p,max} = 1s$ and minimum dwell time of $t_{min} = 0.25s$.

The cost function takes a vector representing the dt for each switching moment as its input. For instance, with two switching instances, the vector could be $[0.25, 0.35]$. This implies the signal begins at time 0.0, transitions to the next dynamic at $0.0 + 0.25 = 0.25$, and then switches again at $0.25 + 0.35 = 0.6$. This gives us the signal sequence $[0.0, 0.25, 0.6]$.

In the Buck-Boost converter example, the initial values for the switching vector are represented by $[0.25, 0.25]$. The lower bounds are defined as $[0.25, 0.25]$ and the upper bounds are set to $[1.5, 1.5]$. The solution provided by the FMINCON optimization yields

$$\begin{gathered} \tau^\infty = [0, 0.2513, 0.5914], \\ \boxed{\Omega^\infty = [1, 2].} \end{gathered} \tag{2.17}$$

The outcome under this condition is displayed in Figure (2.2), showcasing the cyclic trajectory around the average value $[2, -1]$. The corresponding command signal can be seen in Figure (2.4).

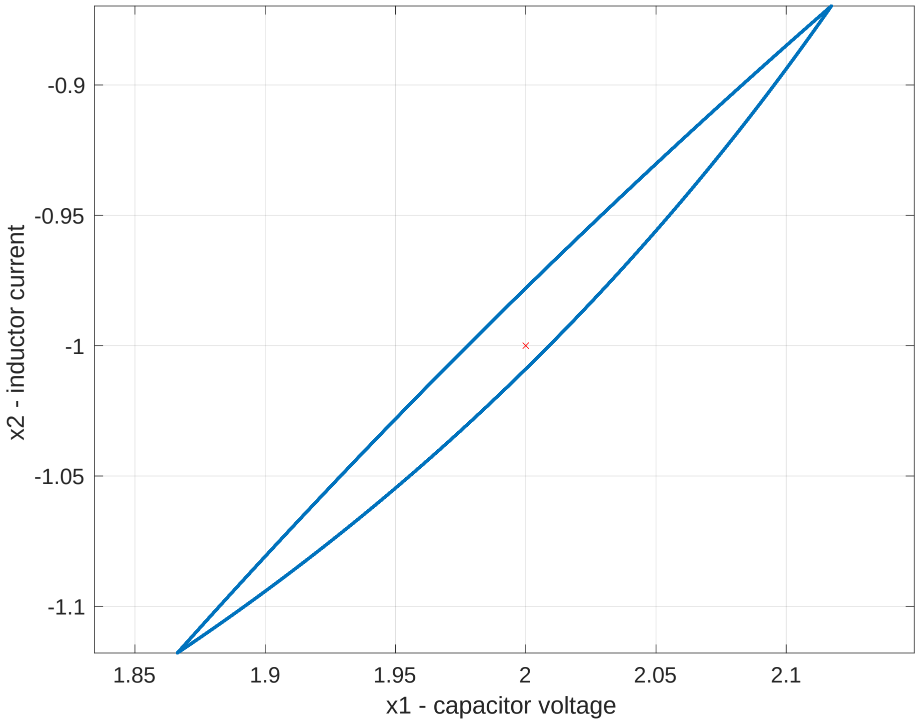

FIGURE 2.3 – Buck-Boost converter: State trajectory

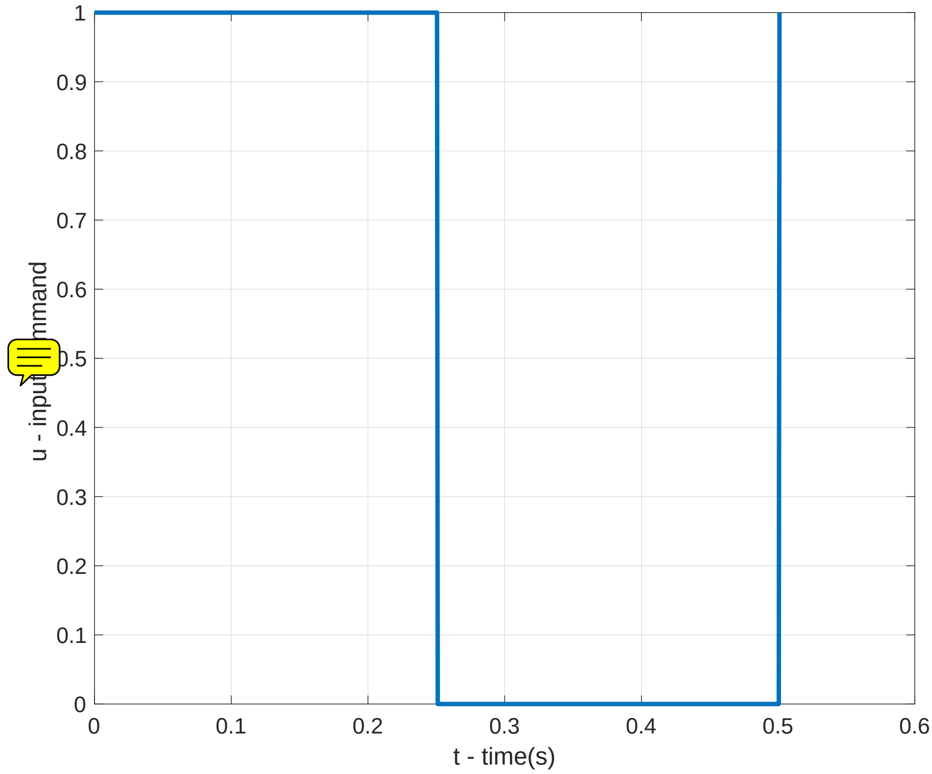

FIGURE 2.4 – Buck-Boost converter: Command signal

2.3.2 Multilevel converter

Considering a Multilevel converter whose topology is shown in Figure 2.5.

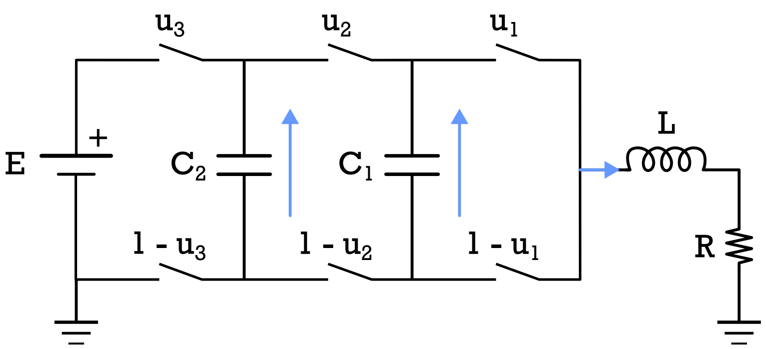

FIGURE 2.5 – Four-level three-cell converter

The state variables are the voltage on the capacitors $x(t) = [v_{c_1}(t), v_{c_2}(t), i_L(t)]$. Where v_{c_1} , v_{c_2} are the voltages in each capacitor and i_L is the load current.

The dynamic of this system can be described as

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \tag{2.18}$$

rearranging the terms we have

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{u_2-u_1}{C1} \\ 0 & 0 & \frac{u_3-u_2}{C2} \\ \frac{u_1-u_2}{L} & \frac{u_2-u_3}{L} & \frac{-R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}u_3 \end{bmatrix} \tag{2.19}$$

Each combination of the switches determines a different behavior

$$\mathbf{u_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, A_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, b_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.20}$$

$$\mathbf{u_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_1} \\ 0 & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.21}$$

$$\mathbf{u_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, A_3 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{-1}{C_2} \\ \frac{-1}{L} & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.22}$$

$$\mathbf{u_4} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, A_4 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, b_4 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.23}$$

$$\mathbf{u_5} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, A_5 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & 0 \\ \frac{1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, b_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.24}$$

$$\mathbf{u_6} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, A_6 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, b_6 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.25}$$

$$\mathbf{u_7} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, A_7 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{C_1} \\ 0 & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, b_7 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.26}$$

$$\mathbf{u_8} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, A_8 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, b_8 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.27}$$

With the detailed dynamic for each switching position, we can define the list of elements for the multilevel converter as $\left\{A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8\right\}$ and $\left\{b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8\right\}$.

The criteria for the optimization is to compute the trajectory around the average value $\left[\frac{2}{3}E, \frac{1}{3}E, 1\right]$ with $E = 30V$. The maximum cycle period is $T_{p,max} = 0.4ms$ and minimum tweel time of $t_{min} = 0.022ms$.

The simulation parameters are $C_1 = 40\mu F$, $C_2 = 40\mu F$, $L = 10mH$ and $R = 10\Omega$ with the weight matrix $Q = diag\left[10, 5, 20000\right]$.

Using the optimal switching sequence

$$\Omega^{\infty} = \left\{1, 2, 4, 8, 3, 1, 5, 8, 5\right\} \tag{2.28}$$

The optimal switching instant is

$$\tau^{\infty} = \left\{0, 0.088, 0.110, 0.132, 0.154, 0.176, 0.198, 0.220, 0.242, 0.264\right\} \tag{2.29}$$

with the initial condition $x(0) = \left[21.71, \boxed{3.29,} 1.08\right]^T$.

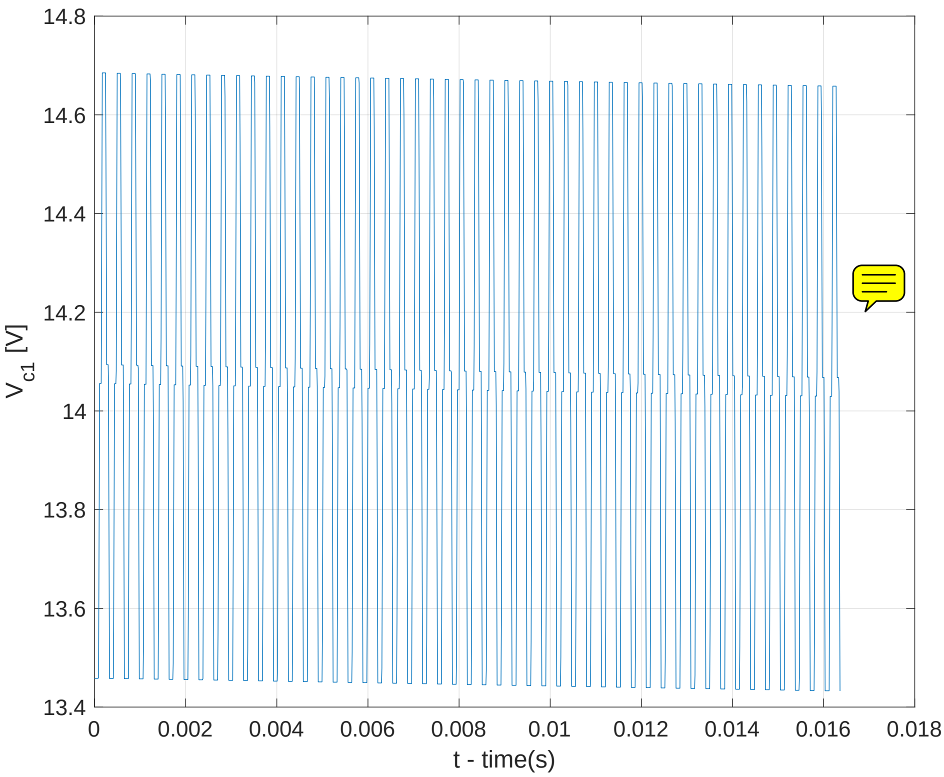

FIGURE 2.6 – Variant Dynamic Circuit 2: Voltage C_1

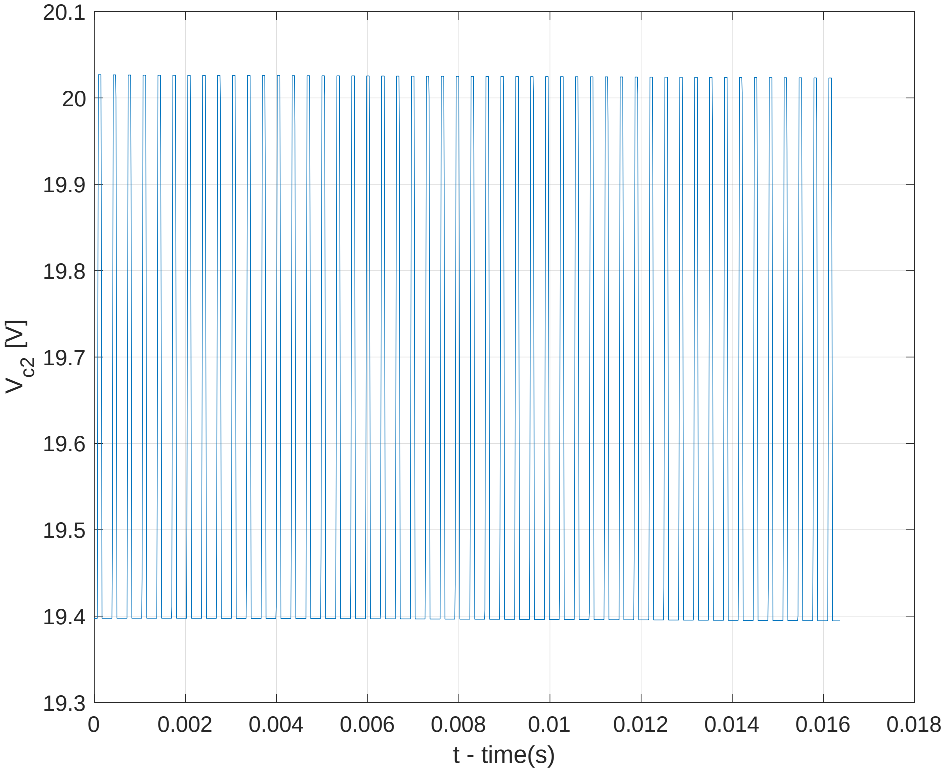

FIGURE 2.7 – Variant Dynamic Circuit 2: Voltage C_2

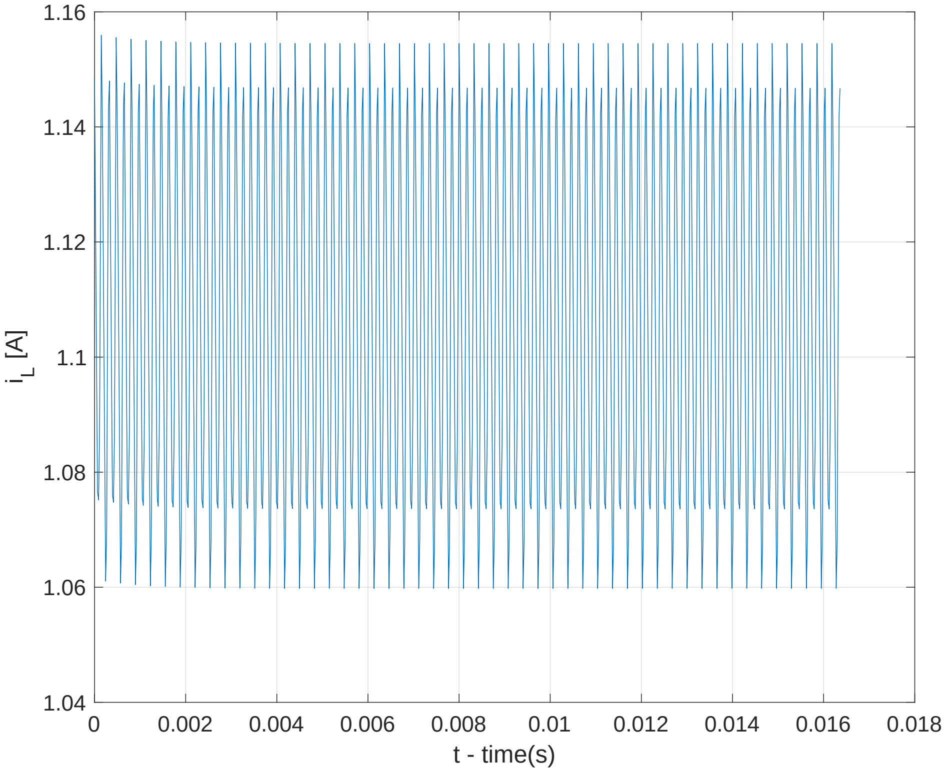

FIGURE 2.8 – Variant Dynamic Circuit 2: Current L

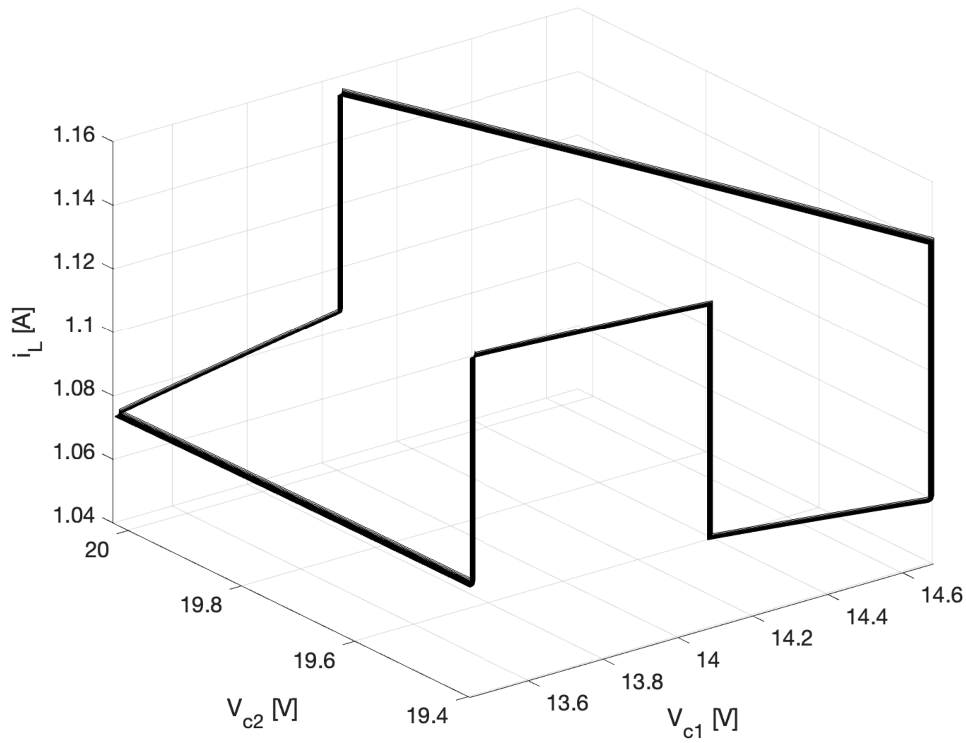

FIGURE 2.9 – Variant Dynamic Circuit 2: State Trajectory